



VICTORIA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
2022
HIGHER 2

NAME:

CT CLASS:

BIOLOGY

Paper 1 Multiple Choice

9744 / 01

23/09/2022

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name and class on the Answer Sheet in the spaces provided.

WRITE and SHADE your EXAM NUMBER on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

- 1** Four students were asked to match the function with the appearance of some cell structures in an animal cell.

The functions were listed by number.

- 1 mRNA passes through to the ribosome
- 2 synthesis of lipids
- 3 packaging of hydrolytic enzymes that will remain in the cell

The appearances were listed by letter.

- V membrane enclosing an inner cavity
- W non-membrane-bound, spherical structures
- X a double membrane interspersed with pores
- Y non-membrane-bound, cylindrical structures
- Z membrane-bound sacs, arranged as a flattened stack

Which student correctly matched the numbered function with the appearance of the cell structure?

	1	2	3
A	V	Z	W
B	W	V	Z
C	X	Y	V
D	X	V	Z

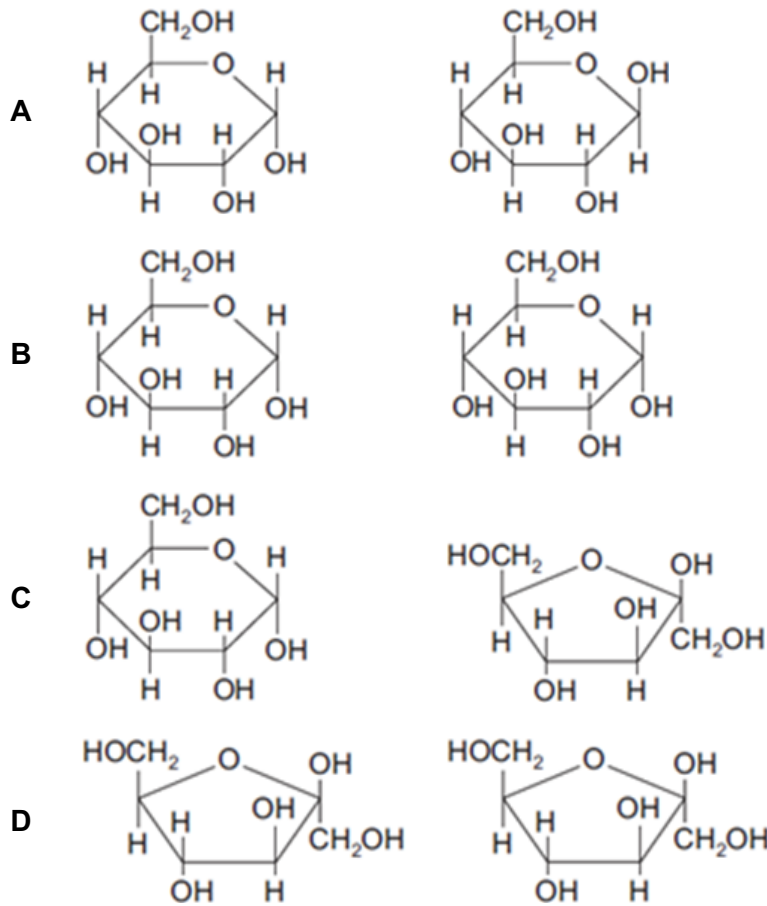
- 2** A scientist observed the following under the microscope.

- 1 Heart muscle cells contract in a synchronised manner.
- 2 Mature red blood cells do not have nuclei.
- 3 Spermatogonia undergo mitotic cell division.
- 4 Tissue samples from endocrine glands contain many cells of different shapes and sizes.

Which of his observations support the cell theory?

- A** 1 and 2
- B** 3 and 4
- C** 1 and 3
- D** 2 and 4

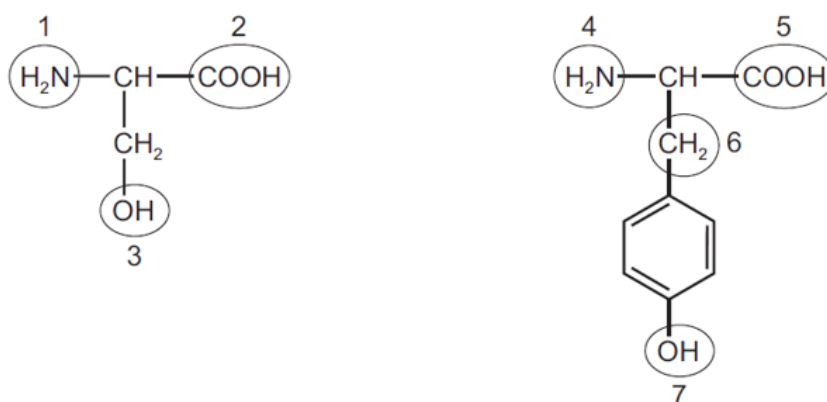
3 Which pair of monosaccharides form a non-reducing sugar?



4 Which of the following is true about triglycerides and phospholipids?

- A** They may have either saturated or unsaturated fatty acids.
- B** A triglyceride has a glycerol but a phospholipid has a phosphate group instead of a glycerol.
- C** All triglycerides have saturated fatty acids while phospholipids have unsaturated fatty acids to maintain membrane fluidity.
- D** They are formed through hydrolysis reactions.

- 5 The diagram shows the structures of two amino acids.

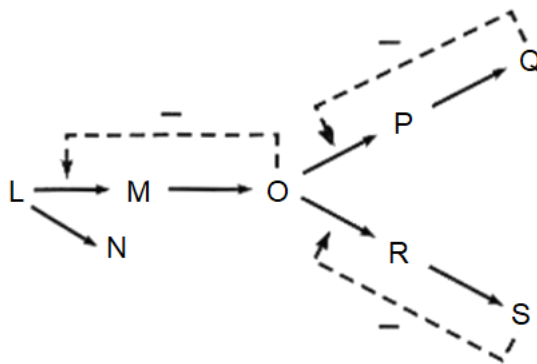


These amino acids will form part of a protein that has a tertiary structure.

Which numbered groups could form bonds to maintain the tertiary structure of the protein?

	numbered groups	type of bond
A	2 and 4	peptide bond
B	1 and 5	hydrogen bond
C	3 and 6	hydrophobic interaction
D	3 and 7	hydrogen bond

- 6 In the following branched metabolic pathway, a dotted arrow with a minus sign symbolises inhibition of a metabolic step by an end product.

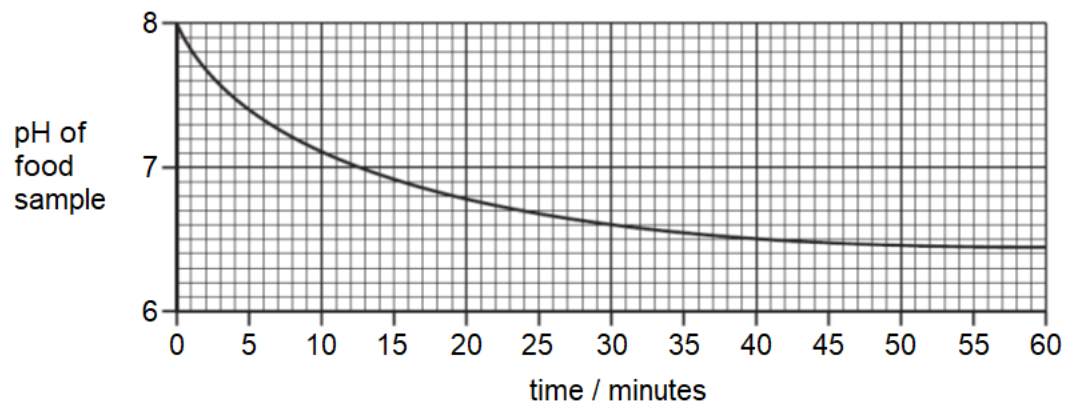


Which reaction would continue if both Q and S were present in the cell in high concentrations?

- A** $L \rightarrow M$
- B** $M \rightarrow O$
- C** $R \rightarrow S$
- D** $L \rightarrow N$

- 7 Lipase is an enzyme that catalyses the hydrolysis of lipids. An experiment was carried out to investigate changes in pH when lipase is added to a food sample containing a high proportion of lipids.

The results are shown in the graph.



Which statements are possible explanations of the results of the experiment between 50 minutes and 60 minutes?

- 1 Enzyme concentration becomes the limiting factor.
 - 2 Substrate concentration becomes the limiting factor.
 - 3 All the enzyme active sites are occupied.
 - 4 Denaturation of the enzyme by the products takes place.
 - 5 There is end-product inhibition.
- A** 1, 2 and 3
B 1, 4 and 5
C 2, 3 and 4
D 2, 4 and 5

- 8 The diagram below shows a micrograph of a multicellular structure formed several days after fertilisation of an egg by a sperm. A type of stem cell can be extracted from this structure.



Which of the following correctly describe the stem cell that can be extracted from this structure?

- 1 It does not exhibit contact inhibition, thus forms a mass of cells.
 - 2 It is totipotent and is able to differentiate into almost all cell types.
 - 3 It is undifferentiated and does not express tissue-specific structures.
 - 4 It expresses telomerase to maintain the length of telomeres.
 - 5 It is metabolically inactive as it does not serve tissue-specific functions.
- A** 1 and 3 only
B 3 and 4 only
C 1, 2 and 4
D 3, 4 and 5
- 9 The DNA sequence CCAAGAAAGTCGACAAACA codes for the amino acid chain gly-ser-ser-ala-val-cys.

As a result of a single-base substitution, the sequence length of the amino acid chain is shortened from six to two amino acids.

Which base in the DNA sequence was changed by the mutation?

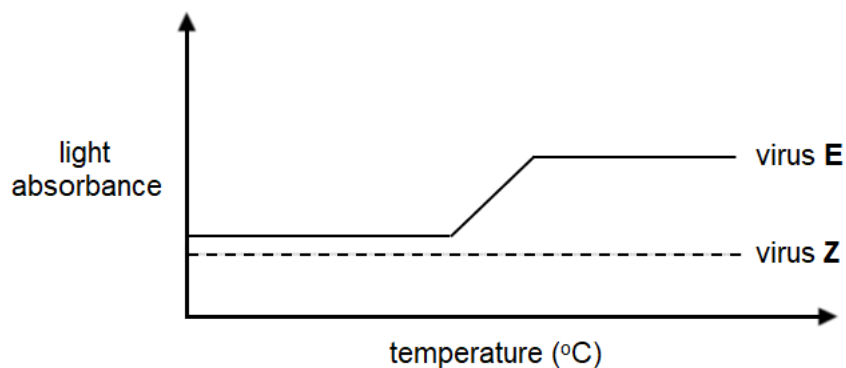
- A** A
B C
C G
D T

10 Which of these steps in protein synthesis involves complementary base pairing?

- A** Amino acid and tRNA form a tRNA-amino acid complex.
- B** A ribosome moves along the mRNA by one codon.
- C** A release factor occupies the ribosome at the stop codon.
- D** Amino acids are correctly aligned in a ribosome.

11 Two new viruses **E** and **Z**, which infect eukaryotic cells, have been identified.

In one experiment, the nucleic acid from each virus is isolated and analysed over a range of temperatures. The light absorbance of nucleic acids changes when denaturation or annealing occurs. The behaviour of the nucleic acid from each virus is shown in the graph.

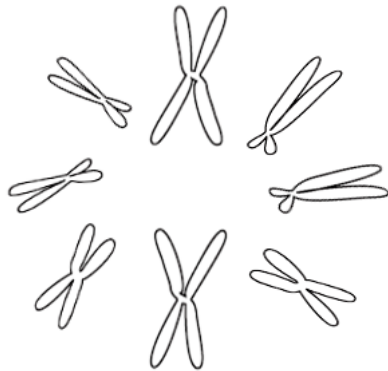


In a second experiment, it is found that treatment with reverse transcriptase inhibitors or with inhibitors of DNA synthesis blocks the ability of virus **Z** to infect cells. In contrast, reverse transcriptase inhibitors have no effect on the ability of virus **E** to infect cells but DNA synthesis inhibitors block infection by virus **E**.

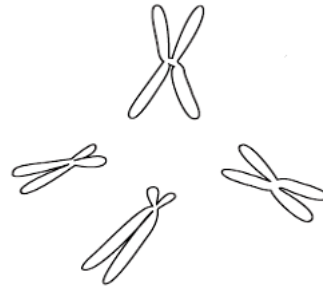
Which of the following conclusions can be drawn from both the experiments?

- A** Genome of virus **E** is single-stranded RNA and that of virus **Z** is double-stranded DNA.
- B** Genome of virus **E** is double-stranded DNA and that of virus **Z** is single-stranded RNA.
- C** Genome of virus **E** is double-stranded RNA and that of virus **Z** is single-stranded DNA.
- D** Genome of virus **E** is double-stranded DNA and that of virus **Z** is double-stranded RNA.

- 12 Figures (A) and (B) below show the chromosomes of a plant cell undergoing two different stages of cell divisions, viewed from the pole.



(A)

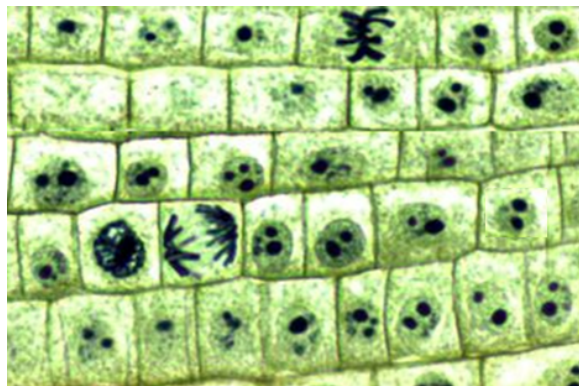


(B)

Which of the following correctly describes the stage of cell division shown in (A) and (B), as well as the diploid number of the plant cell?

	stage of cell division shown in (A)	stage of cell division shown in (B)	diploid number of plant cell
A	prophase	anaphase	4
B	metaphase I	telophase II	8
C	prophase I	prophase II	4
D	metaphase	metaphase II	8

- 13 The diagram below shows a micrograph of the onion root tip.

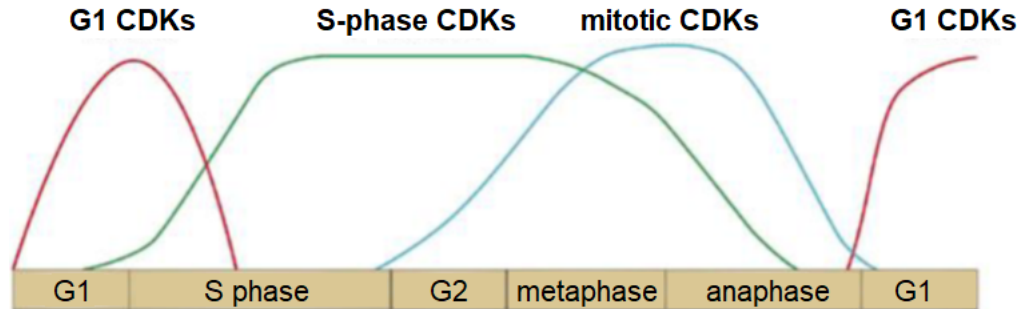


How many of the cells in the diagram are undergoing mitosis?

- A 2
- B 3
- C 20
- D 30

- 14** Cyclins are regulatory proteins that associate with cyclin-dependent kinases (CDKs) to control the different stages of the cell cycle. The right type and amount of cyclins and CDKs must be present at the different stages to ensure regulation of the cell cycle.

The diagram shows the concentrations of the different CDKs.



How could the levels of the different CDKs be regulated during these stages?

- 1 binding of repressor to operator
- 2 formation of heterochromatin
- 3 length of mRNA poly(A) tail
- 4 ubiquitination of CDKs

- A** 1 and 2
B 1 and 3
C 2 and 4
D 3 and 4

- 15** The following statements are about eukaryotic control elements:

- 1 Attachment of RNA polymerase at the TATA box is achieved with the help of a series of specific transcription factors.
- 2 The DNA binding site on general transcription factors and specific transcription factors is the same.
- 3 When the histones found in part of a chromosome are methylated, the control elements of a gene are easily accessed.
- 4 Repressors bind to regions of DNA far upstream of promoters to repress transcription.

Which of the above statement(s) is / are true?

- A** 1 and 3
B 1 and 4
C 2 and 4
D 4 only

- 16** Proteins X and Y play a role in regulating gene expression.

Protein X forms a complex with GTP and mediates the binding of methionyl aminoacyl-tRNA to the small ribosomal subunit which then binds to the 5' end of mRNA and scans for the first AUG codon. When an AUG codon is recognised, protein X hydrolyses bound GTP to GDP and it is released from the small ribosomal subunit. A complete ribosome then forms and protein synthesis begins.

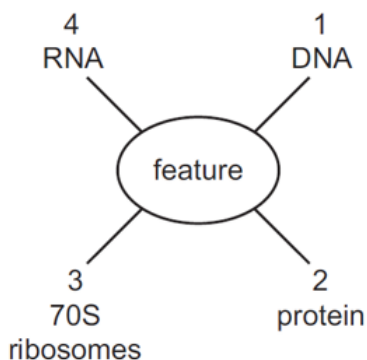
Protein Y is required to cause GDP release from protein X so that it can be reused. The reuse of protein X is inhibited when it is phosphorylated because phosphorylated X binds to protein Y tightly and inactivates protein Y.

Which of the following statements can be concluded?

- 1 Proteins X and Y control gene expression by regulating translational control.
- 2 Protein X is a translational initiation factor that has catalytic activity.
- 3 Inactivation of protein Y will inhibit translation.
- 4 Active protein kinases will decrease overall protein synthesis.

- A** 1, 2, 3 and 4
B 1 and 2 only
C 1 and 3 only
D 2 and 3 only

- 17** The diagram shows some features that occur in organisms.



Which features can be present in viruses?

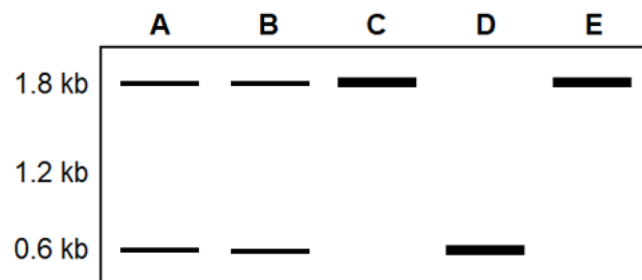
- A** 2 and 4 only
B 1, 2, and 4
C 1, 3 and 4
D 2, 3 and 4

- 18** A family exhibits a rare autosomal recessive disease, which is caused by a point mutation in a gene. This mutation also removes an *EcoR*I restriction site in the gene, such that the *EcoR*I restriction enzyme cannot cleave the DNA at this mutated site.

To ascertain the genotype of family members, researchers did the following:

- 1 amplify a 1.8 kb long DNA sequence containing the gene using PCR
- 2 cut the PCR products with *EcoR*I restriction enzyme
- 3 separate the DNA fragments using an agarose gel
- 4 autoradiography using a radioactively-labelled probe

The result of this experiment is shown in the figure below, using DNA obtained from each of the five family members.



Which of the following shows correctly where the probe binds to, and the family members who have the disease?

	where the probe binds to	family members who have the disease
A		A and B
B		C and E
C		A and B
D		C and E

- 19** Bacterial cells divide by a process called binary fission.

Which macromolecules must be synthesised for binary fission?

- 1 cell membrane proteins and RNA
- 2 DNA and peptidoglycan
- 3 enzymes and cellulose

- A** 1, 2 and 3
- B** 1 and 2 only
- C** 2 and 3 only
- D** 3 only

- 20** A flowering plant has either white, yellow or red flowers. Flower colour is controlled by two independently assorting genes. The dominant allele of gene **A** codes for an enzyme that catalyses the synthesis of a yellow pigment, while the dominant allele of gene **B** codes for another enzyme that converts the yellow pigment into a red pigment. The recessive alleles of both genes code for non-functional enzymes.

A plant with white flowers was crossed with another plant with yellow flowers. All of the F1 progeny plants had red flowers. One of the F1 progeny plants was then back-crossed with the parent plant with white flowers.

What is the probability of obtaining a F2 plant with white flowers?

- A** 25%
- B** 50%
- C** 75%
- D** 100%

- 21** In a genetic study of brinjal plants, a test cross was made between a variety of plant producing purple and long brinjal, and another variety producing green and short brinjal. The results of the F1 generation are shown below.

phenotypes	number
purple, long	28
purple, short	30
green, long	26
green, short	34
total number	118

The formula for chi-squared test and table of probabilities are shown below:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

key:

Σ = 'sum of'

O = observed values

E = expected values

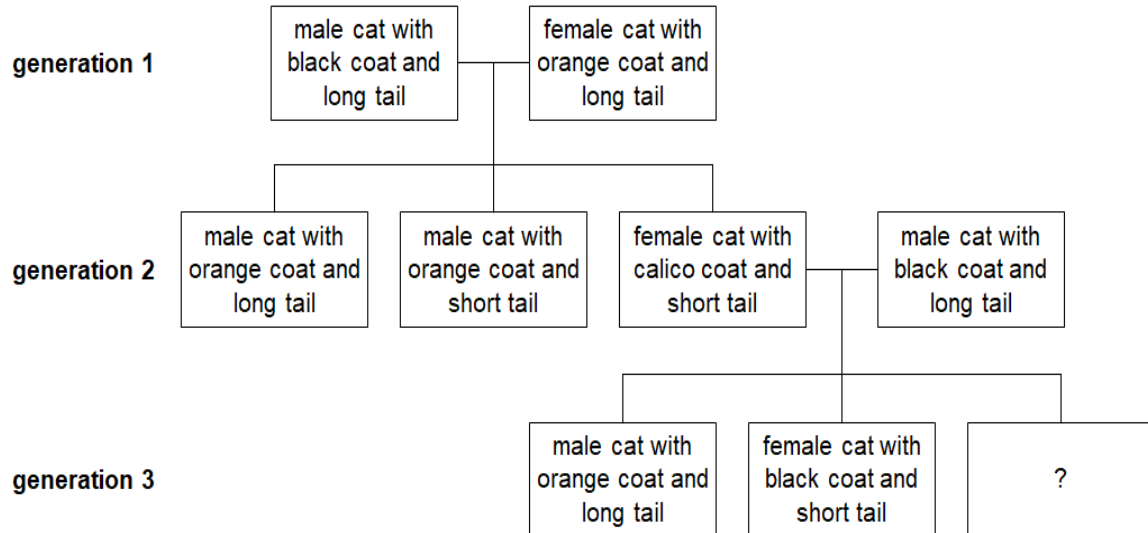
degree of freedom	probability				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Which of the following statements is false?

- A** Probability that differences between the observed and expected numbers is due to chance is greater than 10%.
- B** The two genes coding for the colour and shape are not linked.
- C** Differences between the observed and expected numbers are not statistically significant.
- D** The calculated χ^2 value is greater than the critical χ^2 value.

- 22** Coat colour in cat is controlled by a gene with two codominant alleles. Male cats are either black or orange in colour. Female cats may be black, orange or have both colours (known as calico). Tail length in cat is controlled by another gene with dominant-recessive alleles.

The pedigree below shows a family of cats with different coat colours and tail lengths.



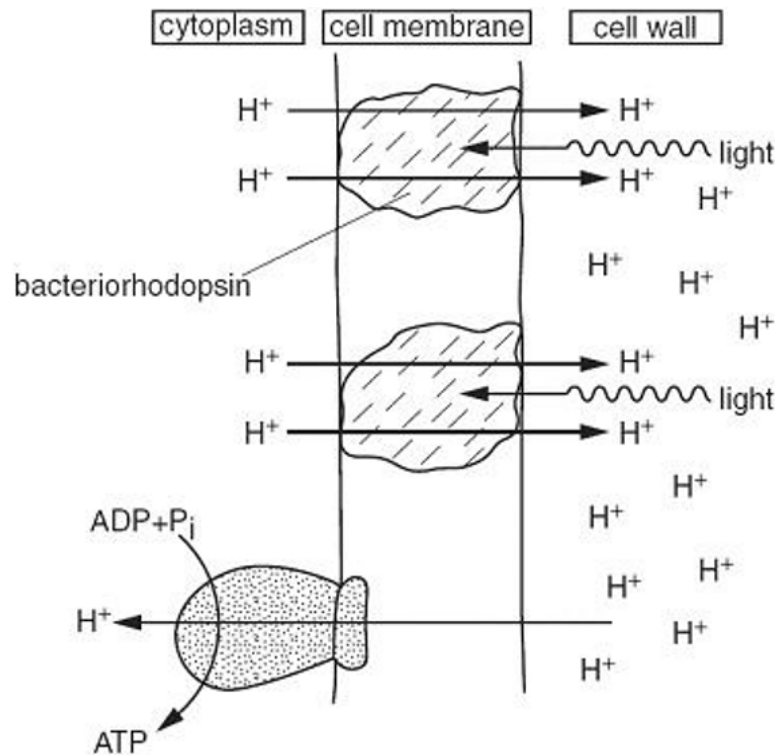
What is the probability that the generation 3 offspring will be a female cat with calico coat and long tail?

- A** 0.125
 - B** 0.25
 - C** 0.5
 - D** 0.625
- 23** Mature red blood cells lack organelles such as nuclei and mitochondria.

Which of the following processes occur during respiration in mature red blood cells?

- A** hydrolysis of glucose
- B** synthesis of nucleotides such as ATP
- C** oxidation of FAD
- D** reduction of oxygen

- 24 The figure illustrates the process of chemiosmosis in a recently discovered photosynthetic bacterium in the Dead Sea.



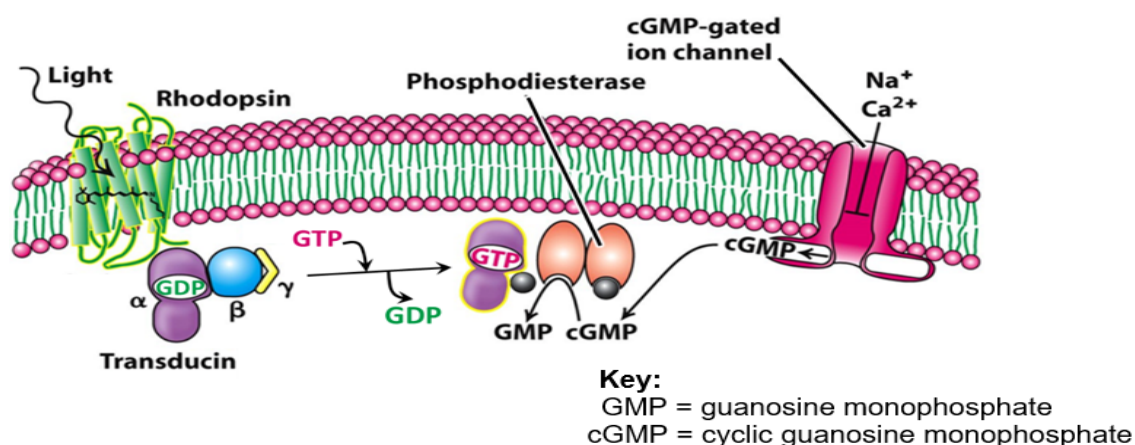
Which of the following statements can be concluded about this photosynthetic bacterium?

- 1 Bacteriorhodopsin acts as a light harvesting complex to harness the energy from the sun.
- 2 The ATP formed in the cytoplasm will be used by the bacterium for cellular activities.
- 3 Protons diffuse down their concentration gradient from the cytoplasm to the cell wall for the eventual production of ATP.
- 4 H^+ diffuses down the concentration gradient back into the cytoplasm via ATP synthase complex. This results in the phosphorylation of ADP to ATP.

- A** 1 and 2 only
B 3 and 4 only
C 1, 2 and 4
D 1, 3 and 4

- 25 Vision is based on the absorption of light by photoreceptor cells in the eye. Detection of light by photoreceptor cells is mediated by a transmembrane receptor protein, rhodopsin. Absorption of light by rhodopsin initiates a cascade of events that closes an ion-channel, causing a change in voltage (difference in charges) across the cell membrane, producing a signal which is communicated to the brain.

The figure below illustrates the signalling events that take place in a photoreceptor cell upon light stimulation.



Which of the following correctly describes the role of the proteins in rhodopsin signalling?

	rhodopsin	transducin	phosphodiesterase	cGMP-gated ion channel
A	a G-protein linked receptor which changes conformation upon light absorption	a G-protein which is activated when the bound GDP replaced by GTP	activated by GTP-bound transducin and converts cGMP to GMP to terminate the transduction	closes when cGMP dissociates, preventing ions from entering the photoreceptor cell
B	a G-protein which changes conformation upon light absorption	a relay protein which is activated when the bound GDP replaced by GTP	converts cGMP to GMP, which is a second messenger that brings about a response	closes when cGMP dissociates, preventing ions from entering the photoreceptor cell
C	a G-protein linked receptor which changes conformation upon binding to G protein	a G-protein which is activated when the bound GDP is phosphorylated to GTP	activated by GTP-bound transducin and converts cGMP to GMP to terminate the transduction	opens when cGMP dissociates, allowing ions to enter the photoreceptor cell
D	a G-protein which changes conformation upon binding to G protein	a relay protein which is activated when the bound GDP is phosphorylated to GTP	converts cGMP to GMP, which is a second messenger that brings about a response	opens when cGMP dissociates, allowing ions to enter the photoreceptor cell

- 26 The chart below shows the taxonomic classification of a particular horse species, *Equus caballus*.

Animalia → Chordata → Mammalia → Perissodactyla → Equidae → *Equus* → *Caballus*

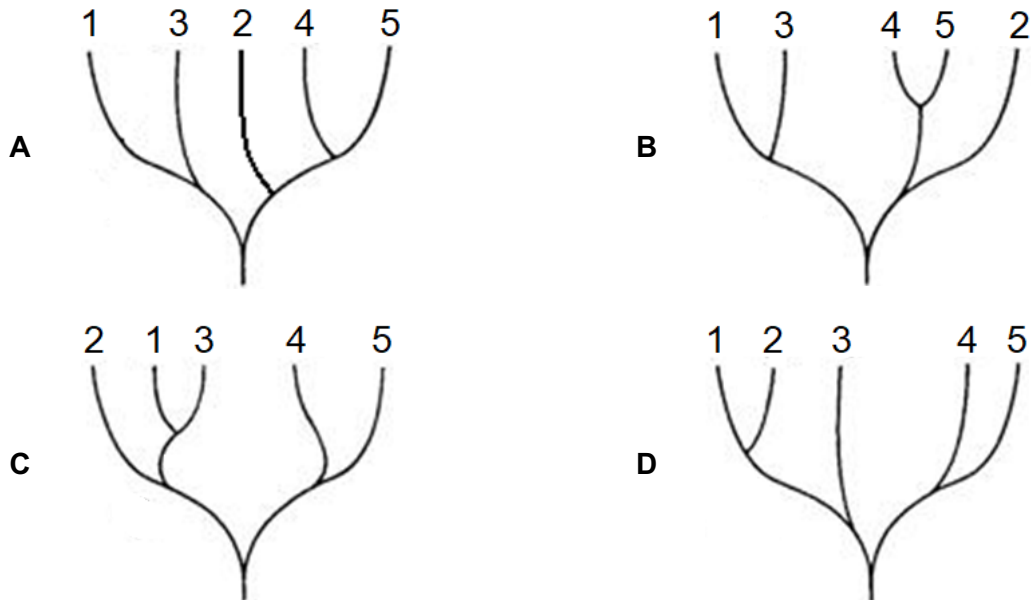
Identify the Family and Class taxa of the horse.

	Family	Class
A	Chordata	Equidae
B	Chordata	Mammalia
C	Equidae	Mammalia
D	Perissodactyla	Equidae

- 27 The table below shows a sequence of 22 amino acid residues of cytochrome c protein from five different species.

species	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
1	G	D	V	E	K	G	K	K	I	F	V	Q	K	C	A	Q	C	H	T	V	E	K
2	G	D	I	E	K	G	K	K	I	D	V	Q	K	C	S	Q	C	H	T	V	E	K
3	G	D	V	E	K	G	K	K	I	F	V	Q	K	C	A	Q	C	H	T	V	E	N
4	G	N	E	D	A	G	A	K	L	F	K	Q	R	C	A	Q	C	H	T	V	Q	A
5	G	N	P	D	A	G	A	K	L	F	K	Q	C	C	A	Q	C	H	T	V	D	A

Using the information above, a suitable phylogenetic diagram would be



28 The following statements describe different cells of the immune system.

- 1 Express cell-surface receptor.
- 2 Release inflammatory cytokines.
- 3 Mature in the thymus.
- 4 Each cell recognises a different antigen.
- 5 Are derived from haematopoietic stem cells.
- 6 Cell functions need to be activated.
- 7 Express MHC class I molecules.
- 8 Actively undergo phagocytosis.

How many of the above statements relate only to macrophage, helper T cell or both?

	number of statements relating to macrophage	number of statements relating to helper T cell	number of statements relating to both cells
A	1	3	4
B	2	2	4
C	2	3	3
D	3	2	3

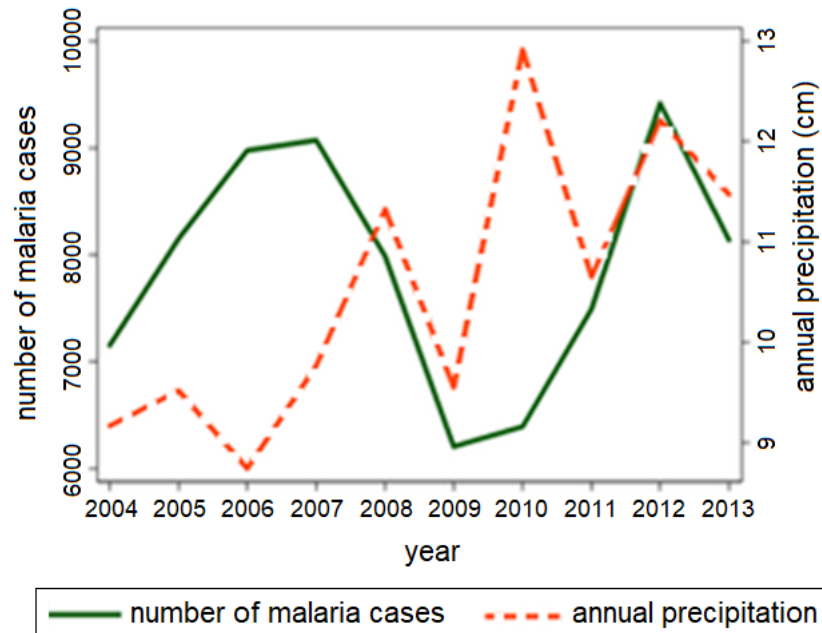
29 While greenhouse effect is a natural phenomenon, today's climate change is a concern.

Which statements support the need to be concerned about today's climate change?

- 1 A small increase in greenhouse gas concentration can have a large effect of increased warming.
- 2 Humans have altered a natural process and exaggerated changes that might normally occur over millions of years.
- 3 Once released into the atmosphere, greenhouse gases remain potent for many years, making it difficult to reverse the process.
- 4 Abrupt changes to the climate system may have unintended outcomes that pose challenges for societies, like more extreme weather, spread of diseases, a decline in marine life, or an alteration of ocean circulation patterns.

- A** 1, 2, 3 and 4
B 1, 2 and 4 only
C 2 and 3 only
D 2 and 4 only

- 30 The graph below shows the relationship between malaria cases and precipitation from 2004 to 2013 of a certain African country.



Which of the following conclusions can be made from the graph?

- 1 Increases in temperature, humidity, and rainfall (all caused by climate change) are helping to proliferate the mosquito population at higher altitudes.
- 2 The geographical distribution of the disease widened, allowing it to emerge in new locations which had previously not supported mosquito populations.
- 3 Increases in temperatures at lower altitudes, where mosquitoes and malaria are already prevalent, have the impact of adjusting the growth cycle of the parasite that causes the mosquito to carry the disease, allowing it to develop malaria faster and, therefore, increasing transmission rates.
- 4 Additional rainfall that can result from changes to the El Niño cycle can provide good breeding conditions in locations that are usually dry, allowing the insect and therefore, the disease to proliferate in new regions.
- 5 Droughts caused by changes to weather patterns in conjunction with increased humidity can also result in the proliferation of the mosquito populations. In this case, rivers can be converted into strings of pools which provide the optimal breeding site for mosquitoes.
- 6 Changes to the El Niño cycle that cause areas that are significantly impacted by malaria to become drier can lead to increases in malaria cases due to the loss of heavy rainfall that can wash away mosquito breeding sites.

- A** 1, 2, 3 and 4
B 1, 2 and 4 only
C 3, 4 and 5 only
D 4, 5 and 6 only